Import Libraries:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from scipy import stats
import statsmodels.api as sm
from statsmodels.formula.api import ols
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
```

Upload Dataset

```
from google.colab import files

uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving titanic.csv to titanic.csv

```
df = pd.read_csv("titanic.csv")
print(df.head())
```

```
   PassengerId  Survived  Pclass  \
0          892         0       3
1          893         1       3
2          894         0       2
3          895         0       3
4          896         1       3

                                          Name     Sex   Age  SibSp  Parch  \
0                              Kelly, Mr. James    male  34.5      0      0
1              Wilkes, Mrs. James (Ellen Needs)  female  47.0      1      0
2                     Myles, Mr. Thomas Francis    male  62.0      0      0
3                              Wirz, Mr. Albert    male  27.0      0      0
4  Hirvonen, Mrs. Alexander (Helga E Lindqvist)  female  22.0      1      1

    Ticket     Fare Cabin Embarked
0   330911   7.8292   NaN        Q
1   363272   7.0000   NaN        S
2   240276   9.6875   NaN        Q
3   315154   8.6625   NaN        S
4  3101298  12.2875   NaN        S
```

Descriptive statistics

```
#Numerical columns
print("Mean:")
print(df[["Age", "Fare"]].mean())

print("\nMedian:")
print(df[["Age", "Fare"]].median())

print("\nMode:")
print(df[["Age", "Fare"]].mode().iloc[0])

print("\nVariance:")
print(df[["Age", "Fare"]].var())

print("\nStandard Deviation:")
print(df[["Age", "Fare"]].std())
```

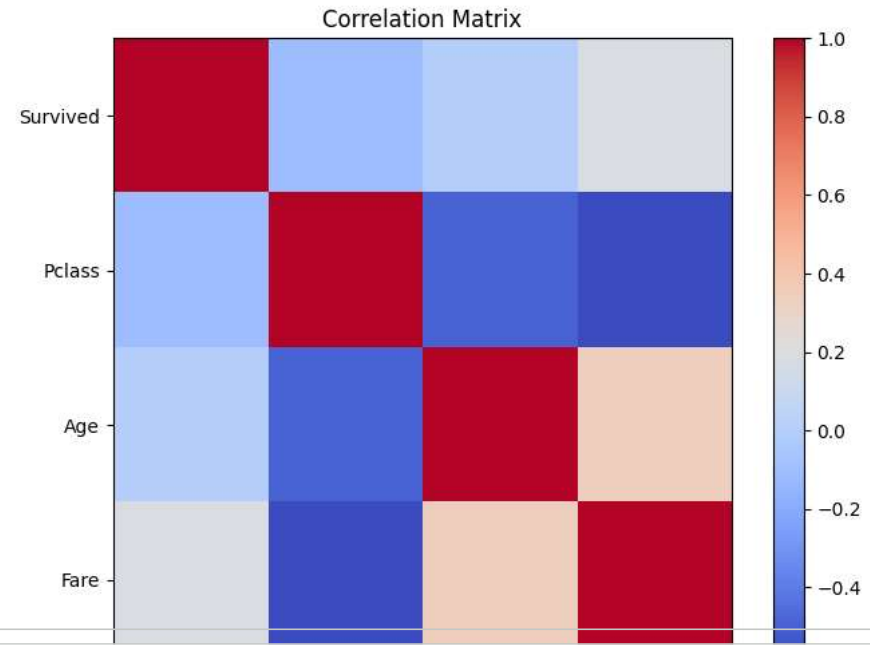
```
Mean:
Age     30.272590
Fare    35.627188
dtype: float64

Median:
Age     27.0000
```

```
Fare    14.4542
dtype: float64

Mode:
Age     21.00
Fare     7.75
Name: 0, dtype: float64

Variance:
Age      201.106695
Fare    3125.657074
dtype: float64

Standard Deviation:
Age     14.181209
Fare    55.907576
dtype: float64
```

Pearson correlation

```
# Correlation between numerical variables
correlation = df[["Survived", "Pclass", "Age", "Fare"]].corr()

print(correlation)

# Correlation matrix visualization
plt.figure(figsize=(8, 6))

plt.imshow(correlation, cmap="coolwarm", interpolation="none")
plt.colorbar()

plt.xticks(range(len(correlation.columns)), correlation.columns)
plt.yticks(range(len(correlation.columns)), correlation.columns)

plt.title("Correlation Matrix")
plt.show()
```

```
          Survived    Pclass       Age      Fare
Survived  1.000000 -0.108615 -0.000013  0.191514
Pclass   -0.108615  1.000000 -0.492143 -0.577147
Age      -0.000013 -0.492143  1.000000  0.337932
Fare      0.191514 -0.577147  0.337932  1.000000
```




```
# Remove missing Fare values
survived_fare = df[df["Survived"] == 1]["Fare"].dropna()
not_survived_fare = df[df["Survived"] == 0]["Fare"].dropna()

# Independent sample t-test
t_stat, p_value = stats.ttest_ind(
    survived_fare,
    not_survived_fare,
    equal_var=False
```

```
)

print("T-statistic:", t_stat)
print("P-value:", p_value)

if p_value < 0.05:
    print("Reject H0: There is a significant difference in Fare.")
else:
    print("Fail to reject H0: There is no significant difference in Fare.")
```

```
T-statistic: 3.4478682983060884
P-value: 0.0006845156351814127
Reject H0: There is a significant difference in Fare.
```

One-Way ANOVA

```
# Create groups based on passenger class
class1 = df[df["Pclass"] == 1]["Fare"].dropna()
class2 = df[df["Pclass"] == 2]["Fare"].dropna()
class3 = df[df["Pclass"] == 3]["Fare"].dropna()

# One-Way ANOVA
f_stat, p_value = stats.f_oneway(class1, class2, class3)

print("F-statistic:", f_stat)
print("P-value:", p_value)

if p_value < 0.05:
    print("Reject H0: There is a significant difference among passenger classes.")
else:
    print("Fail to reject H0: No significant difference among passenger classes.")
```

```
F-statistic: 129.89190776865902
P-value: 1.641767321164957e-44
Reject H0: There is a significant difference among passenger classes.
```

simple Linear Regression mode

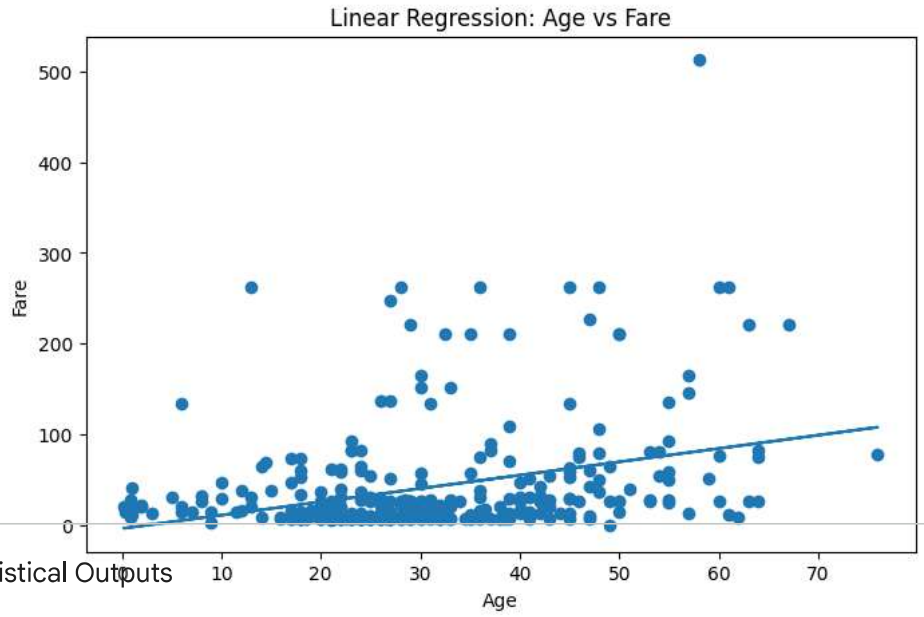
```
# Select required columns
reg_data = df[["Age", "Fare"]].dropna()

# Independent variable
X = reg_data["Age"]

# Dependent variable
y = reg_data["Fare"]

# Add constant
X = sm.add_constant(X)

# Build regression model
model = sm.OLS(y, X).fit()

# Display results
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                   Fare   R-squared:                       0.114
Model:                            OLS   Adj. R-squared:                  0.112
Method:                 Least Squares   F-statistic:                     42.41
Date:                Mon, 07 Sep 2026   Prob (F-statistic):           2.76e-10
Time:                        08:44:10   Log-Likelihood:                -1811.0
No. Observations:                 331   AIC:                             3626.
Df Residuals:                     329   BIC:                             3634.
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         -3.2931      7.502     -0.439      0.661     -18.051      11.465
Age            1.4670      0.225      6.513      0.000       1.024       1.910
==============================================================================
Omnibus:                      241.179   Durbin-Watson:                   2.017
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             2693.317
Skew:                           3.014   Prob(JB):                         0.00
```

```
Kurtosis:                       15.607   Cond. No.                         78.8
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Regression Graph

```
plt.figure(figsize=(8, 5))

plt.scatter(reg_data["Age"], reg_data["Fare"])

# Prediction line
predicted = model.predict(X)

plt.plot(reg_data["Age"], predicted)

plt.xlabel("Age")
plt.ylabel("Fare")
plt.title("Linear Regression: Age vs Fare")

plt.show()
```



Statistical Outputs

```
print("Regression Coefficients:")
print(model.params)

print("\nP-values:")
print(model.pvalues)

print("\nConfidence Intervals:")
print(model.conf_int())

print("\nR-squared:")
print(model.rsquared)
```

```
Regression Coefficients:
const   -3.293097
Age      1.466976
dtype: float64

P-values:
const    6.609779e-01
Age      2.763372e-10
dtype: float64

Confidence Intervals:
                0          1
const -18.051091  11.464897
Age     1.023864   1.910087

R-squared:
0.11419775583782077
```